

Postmagmatic & Deuteric Alteration: Saussuritization of Plagioclase, Uralitization, Chloritization & Devitrification

Comprehensive Petrology & Mineralogy Guide · Authoritative References: John D. Winter (2010, Ch. 3 & 26) & Dexter Perkins (2014, Ch. 13)

THEORETICAL FOUNDATION: DEUTERIC & AUTOMETAMORPHIC PROCESSES

Magmatic Volatiles, Sub-solidus Hydration Reactions, Plagioclase Breakdown & Reaction Stoichiometry

1. The Concept of Deuteric Alteration (Autometamorphism)

During the final stages of igneous crystallization, magma bodies cool through sub-solidus temperatures. Primary, anhydrous high-temperature igneous minerals (calcic plagioclase, olivine, pyroxenes) become thermodynamically unstable in the presence of water-rich residual fluids exsolved from the magma itself:

Deuteric Alteration (Autometamorphism): The alteration of early-formed igneous minerals by aqueous, volatile-rich fluids that exsolved directly from the *same magma body* as it cooled through late magmatic and post-solidus regimes (500°C to 200°C). It is strictly distinguished from external hydrothermal alteration or regional metamorphism, which involve fluids or tectonic stresses from outside sources.

*"Secondary mineral reactions that occur in igneous rocks as they cool, and that are not products of a distinct later metamorphic event, are commonly called **autometamorphic processes** rather than metamorphic ones because they are a natural part of igneous cooling. Autometamorphic processes are more common in plutonic rocks than in volcanics because they remain at elevated temperatures for a longer time."*

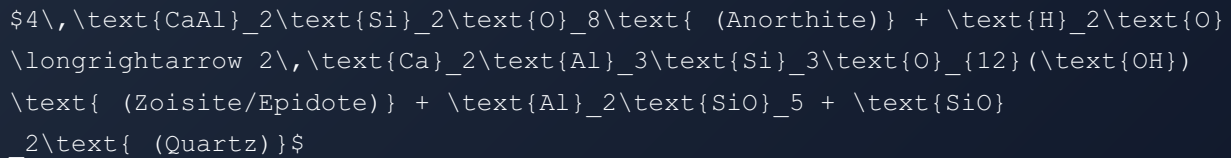
— John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., 2010), Chapter 3: Textures of Igneous Rocks, p. 51.

2. Saussuritization: The Decomposition of Calcic Plagioclase

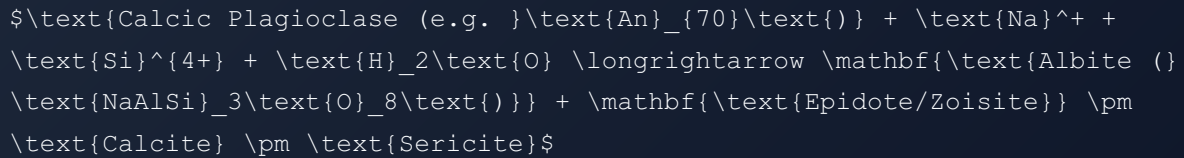
Calcic plagioclase (labradorite, bytownite, anorthite $\text{CaAl}_2\text{Si}_2\text{O}_8$) is an anhydrous framework aluminosilicate that crystallizes at high temperatures ($>1000^{\circ}\text{C}$). Under deuterically hydrated, low-temperature conditions, the anorthite component cannot survive:

THE STOICHIOMETRIC SAUSSURITIZATION REACTIONS

Hydration Breakdown of Anorthite to Zoisite/Epidote:



Coupled Breakdown with Sodium Influx (Spilitization / Saussuritization):



Under the petrographic microscope, saussuritization produces a diagnostic appearance: the originally clear, transparent plagioclase laths turn turbid, cloudy, or "dusty" in plane-polarized light (PPL), resolving under high magnification into a dense microcrystalline felt of **high-relief, brightly polarizing epidote/clinozoisite grains embedded in clear, low-relief albite**.

*"Plagioclase is commonly altered by deuteric or hydrothermal fluids. Calcic plagioclase in particular is unstable at low temperatures in the presence of water, breaking down to an intimate, fine-grained mixture of **albite and epidote (or zoisite/clinozoisite)**, often with variable amounts of sericite (fine muscovite) and calcite. This fine-grained replacement aggregate is termed **saussurite**, after the Swiss geologist H. B. de Saussure."*

— Dexter Perkins, *Mineralogy* (3rd Ed., 2014), Chapter 13: Sorosilicates (Epidote Group), p. 411.

3. Petrographic Evidence: Photomicrographs of Deuteric Alteration

Figure 20 from Winter illustrates how autometamorphic fluids attack primary minerals along grain boundaries and cleavage planes, generating classic secondary replacement textures:

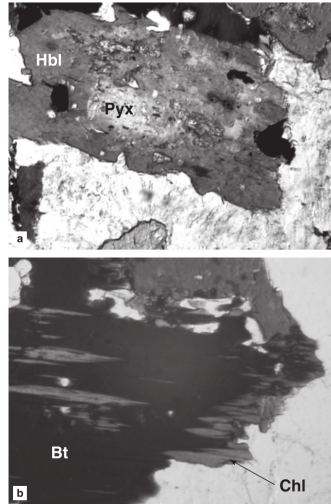


FIGURE 20 (a) Pyroxene largely replaced by hornblende. Some pyroxene remains as light areas (Pyx) in the hornblende core. Width 1 mm. (b) Chlorite (light) replaces biotite (dark) at the rim and along cleavages. Tonalite, San Diego, California. Width 0.3 mm.

generally replaces the initial mafic from the rim inward. In the case of biotite, H_2O may work its way along the prominent cleavages, and chlorite can be seen to replace biotite margins as well as along cleavage planes (Figure 20b).

Sericite is a term applied to any very fine-grained white mica. **Seritization** is the process by which felsic minerals (usually feldspars or feldspathoids in igneous rocks) are hydrated to produce sericite. Incipient stages can be recognized by a fine, dusty appearance of feldspars in plane-polarized light. In more advanced stages of alteration, the feldspars appear speckled, with fine micas having yellowish birefringence and then larger clots with coarser crystals and higher birefringence. K^+ ions are required for plagioclase to be altered to the common forms of sericite. Potassium may be released by chloritization of nearby biotite. K-feldspar requires no additional K^+ and may be more sericitized than associated plagioclase.

Saussuritization is the alteration of plagioclase to produce an epidote mineral. Higher-temperature plagioclase tends to be more calcium-rich, which is less stable than its sodic counterpart at low temperatures. The calcium-rich types thus break down to more nearly pure albite,

Figure 1 (Winter Fig. 20): Thin section photomicrographs of postmagmatic deuteric alterations. (a) **Uralitization:** Primary pyroxene largely replaced by fibrous hornblende/actinolite, with remnant pyroxene surviving as light patches. (b) **Chloritization:** Secondary chlorite (light green) replacing primary igneous biotite (dark brown) along cleavage planes and crystal rims in tonalite, San Diego County, California. (Source: John D. Winter, *Principles of Igneous and Metamorphic Petrology*, 2010, Pearson, Chapter 3).

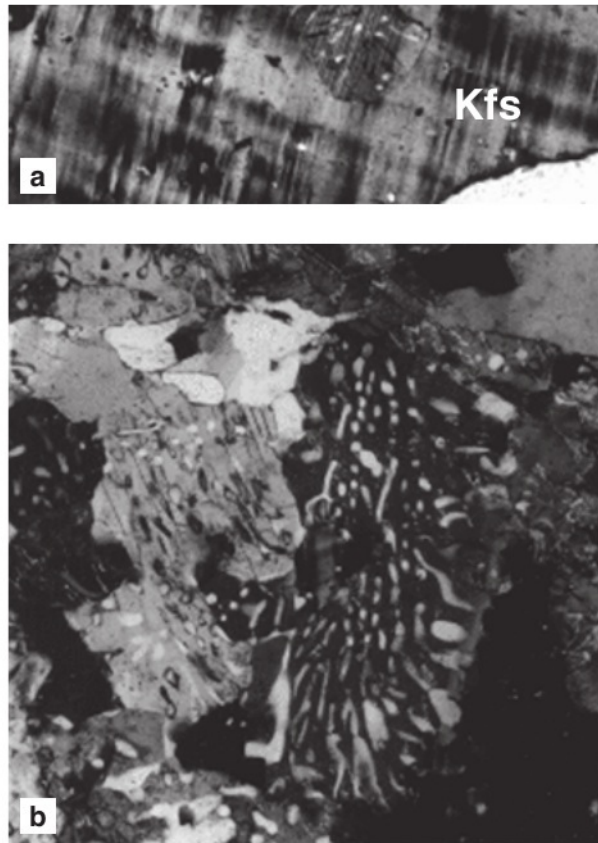


FIGURE 21 Myrmekite formed in plagioclase at the boundary with K-feldspar. Photographs courtesy L. Collins.

Figure 2 (Winter Fig. 21): Late-magmatic / deuteric reaction texture: **Myrmekite** forming in plagioclase at the boundary with K-feldspar, showing vermicular (worm-like) quartz rods intergrown with sodic plagioclase as late fluids re-equilibrate alkali feldspar.

4. The Four Named Alteration Processes Compared

The four options in Question 17 represent the four most important postmagmatic and low-grade alteration processes recognized in igneous petrology. Their distinct reactants, products, and mechanisms are summarized below:

Process Name	Primary Reactant (Starting Material)	Essential Secondary Products	Diagnostic Petrographic Characteristics
Saussuritization ★ Correct Option	Ca-rich Plagioclase (labradorite, bytownite, anorthite)	Albite + Epidote / Zoisite ($\text{CaAl}_2\text{Si}_2\text{O}_8$ sericite, calcite, quartz)	Plagioclase laths turn turbid/cloudy with high-relief epidote granules in clear sodic matrix.
Devitrification Option (b)	Volcanic Glass (obsidian, pitchstone, tachylite)	Quartz + Alkali Feldspar aggregates (spherulites, felsitic chert-like mass, palagonite)	Loss of isotropic glassy character; appearance of radiating spherulitic fibrous fans or microgranular quartz-feldspar.
Chloritization Option (c)	Ferromagnesian Minerals (biotite, hornblende, augite)	Chlorite ($\text{CaAl}_2\text{Si}_2\text{O}_8$ titanite, rutile needles, Fe-Ti oxides)	Green pleochroism, anomalous "Berlin blue" or brown interference colors replacing mafic sheets.
Uralitization Option (d)	Pyroxenes (clinopyroxene: augite; orthopyroxene: hypersthene)	Actinolitic Amphibole ("uralite", fibrous hornblende)	Fibrous, ragged, pale green amphibole pseudomorphs retaining original prismatic pyroxene outlines.

5. Additional Essential Secondary & Deuteric Textures

To ensure total exam readiness, students must also be familiar with related sub-solidus replacement processes in igneous rocks:

- **Sericitization:** Hydration of feldspars (especially K-feldspar and plagioclase) into microscopic flakes of white mica (sericite, $\text{KAl}_3\text{Si}_3\text{O}_{10}(\text{OH})_2$). Imparts a dusty golden-yellow birefringence under crossed polars.
- **Serpentinization:** Hydration of magnesium-rich olivine and orthopyroxene to serpentine ($\text{Mg}_3\text{Si}_2\text{O}_5(\text{OH})_4$) and magnetite: $2\text{Mg}_2\text{SiO}_4 + 3\text{H}_2\text{O} \rightarrow \text{Mg}_3\text{Si}_2\text{O}_5(\text{OH})_4 + \text{Mg(OH)}_2$.
- **Iddingsitization:** Pseudomorphic alteration of olivine to a reddish-brown, semi-opaque mixture of goethite, hematite, smectite, and chlorite (iddingsite).
- **Spilitization:** Sub-seafloor hydrothermal metasomatism of basalt, where calcic plagioclase is altered to albite and augite to chlorite/actinolite, converting basalt into "spilite".



EXAM QUESTION & RIGOROUS OPTION-BY-OPTION DISSECTION

DGMS 2019 Examination · Official Question Statement, Verified Answer & Detailed Distracter Analysis

DGMS 2019 · QUESTION 17

Ca-rich plagioclase breaks down to almost pure albite and epidote during which one of the following deuteric alterations?

- (a) Saussuritization
- (b) Devitrification
- (c) Chloritization
- (d) Uralitization

Correct Option: (a) Saussuritization



Verified via John D. Winter (Igneous & Metamorphic Petrology, pp. 51–55) & Dexter Perkins (Mineralogy, p. 411)

CORRECT ANSWER Option (a): Saussuritization

Detailed Validation: **Saussuritization** is the universally accepted geological term for the sub-solidus deuteric or hydrothermal alteration of calcic plagioclase (labradorite to anorthite) into a fine-grained aggregate of **almost pure albite** and **epidote-group minerals (epidote, zoisite, or clinozoisite)**, often accompanied by minor sericite, calcite, and quartz. The calcium and aluminium released from the anorthite molecule ($\text{CaAl}_2\text{Si}_2\text{O}_8$) combine with hydroxyl ions to crystallize epidote ($\text{Ca}_2\text{Al}_3\text{Si}_3\text{O}_{12}(\text{OH})$), leaving the sodium-silica component behind as albite ($\text{NaAlSi}_3\text{O}_8$). This matches the exam question verbatim.

INCORRECT Option (b): Devitrification

Detailed Refutation: *Devitrification* is the slow, secondary sub-solidus crystallization of amorphous volcanic **glass** into fine-grained crystalline aggregates (typically quartz, cristobalite, and alkali feldspar, forming spherulites or felsitic textures). Devitrification does not involve plagioclase breakdown; it acts exclusively upon non-crystalline volcanic glass.

INCORRECT Option (c): Chloritization

Detailed Refutation: *Chloritization* is the alteration of primary **ferromagnesian (mafic) minerals**—specifically biotite, hornblende, augite, or olivine—into hydrous phyllosilicate minerals of the **chlorite** group ($(\text{Mg,Fe})_5\text{Al}(\text{Si}_3\text{Al})\text{O}_{10}(\text{OH})_8$). Plagioclase contains virtually no magnesium or iron, and cannot break down into chlorite.

INCORRECT Option (d): Uralitization

Detailed Refutation: *Uralitization* is the deuteric or low-grade hydrothermal alteration of **pyroxene** (most commonly augite or pigeonite) into fibrous, pale-green amphibole (actinolitic hornblende or tremolite-actinolite, historically known as "uralite"). It is a pyroxene-to-amphibole chain-silicate transformation, entirely independent of plagioclase feldspar.



SYNTHESIS & HIGH-YIELD PRACTICE QUESTIONS

Curated Problems Covering Deuteric Reactions, Petrography, Metasomatism & Alteration Assemblages

High-Yield Exam Practice Questions

Practice Q1 (Deuteric Mineralogy): Identification of Saussurite

In thin section, a calcic plagioclase phenocryst appears clouded with high-relief, prismatic granules exhibiting high birefringence and parallel extinction, alongside fine flakes with moderate birefringence. What are these two secondary minerals?

Solution & Rationale:

The high-relief grains are **epidote / zoisite** (which concentrate the calcium from anorthite), and the fine, moderate-birefringence flakes are **sericite** (fine-grained white mica, incorporating trace potassium). Together with secondary albite, they constitute saussurite.

Practice Q2 (Glass Alteration): Palagonite vs. Felsite

What is the distinct name given to the devitrification product of basaltic (mafic) volcanic glass formed under aqueous quenching, as opposed to silicic glass?

Answer: Palagonite. Quenched basaltic glass (sideromelane) hydrates and oxidizes into yellowish-orange to brown, optically isotropic palagonite. In contrast, silicic glass devitrifies into microcrystalline quartz-feldspar (felsitic texture or spherulites).

Practice Q3 (Uralite Chemistry): Reactant and Product

Write the simplified mineral reaction showing how augite clinopyroxene is converted to uralite amphibole during postmagmatic cooling.

Answer:

$$5\text{Ca}(\text{Mg,Fe})\text{Si}_2\text{O}_6 \text{ (Augite)} + 2\text{H}_2\text{O} \rightarrow \text{Ca}_2(\text{Mg,Fe})_5\text{Si}_8\text{O}_{22}(\text{OH})_2 \text{ (Actinolite/Uralite)} + 3\text{CaO} + 2\text{SiO}_2$$

The released CaO often combines with surrounding titania and silica to form secondary titanite (sphene).

Practice Q4 (Feldspar Alteration): Sericitization Drivers

Why does plagioclase alter to sericite more readily when associated with biotite that is simultaneously undergoing chloritization?

Answer: Sericite ($\text{KAl}_3\text{Si}_3\text{O}_{10}(\text{OH})_2$) requires potassium (K^+), which is absent in pure plagioclase. When nearby igneous biotite alters to chlorite, it expels its interlayer K^+ into the intergranular fluid, supplying the exact potassium ions necessary to sericitize adjacent plagioclase crystals.

Practice Q5 (Metamorphic Equivalent): Greenschist Facies Analogy

Which regional metamorphic facies exhibits mineral reactions that are thermodynamically and mineralogically identical to saussuritization and chloritization?

Answer: Greenschist Facies (and Sub-greenschist / Prehnite-Pumpellyite facies). In metabasites, the characteristic greenschist assemblage—**Albite + Epidote + Chlorite + Actinolite + Quartz**—is formed by the exact same breakdown of high-temperature basaltic plagioclase + clinopyroxene.

Practice Q6 (Replacement Textures): Myrmekite Definition

Define myrmekite and identify the primary minerals involved in its development.

Answer: Myrmekite is an intergrowth of sodic plagioclase (oligoclase or albite) and vermicular (worm-like) rods of quartz, typically occurring at the boundaries of K-feldspar megacrysts in granitic plutons. It is produced by late-magmatic / deuteric sodium metasomatism displacing potassium: $\text{KAlSi}_3\text{O}_8 + \text{Na}^+ \rightarrow \text{NaAlSi}_3\text{O}_8 + \text{K}^+ + \text{excess SiO}_2$.

Practice Q7 (Sub-seafloor Alteration): Spilitization

Basaltic lavas on the ocean floor that undergo hydrothermal metasomatism with seawater, transforming their calcic plagioclase into albite and olivine/pyroxene into chlorite, are petrologically known as:

Answer: Spilites (process: **Spilitization**). Spilites retain the primary basaltic pillow structures and ophitic/sub-ophitic textures while possessing an entirely sodic, hydrous secondary mineralogy.

Practice Q8 (Zoning & Selective Alteration): Anorthite Susceptibility

In normally zoned plagioclase phenocrysts (calcic core, sodic rim), why does saussuritization selectively attack only the central core of the crystals?

Answer: The core contains the highest concentration of the calcic anorthite molecule ($\text{CaAl}_2\text{Si}_2\text{O}_8$), which has the lowest thermodynamic stability at sub-solidus hydrous temperatures. The sodic rim (albite-rich) is thermodynamically stable in the presence of water at lower temperatures and remains pristine and unaltered.

Comprehensive Secondary Alteration Matrix

Alteration Type	Starting Mineral	Primary Secondary Minerals	Diagnostic Microscopic Appearance	Geological / Geodynamic Context
Saussuritization	Calcic Plagioclase	Albite + Epidote / Zoisite	Cloudy feldspar with microcrystalline high-relief epidote inclusions.	Deuteric plutonic cooling; hydrothermal spilitization.
Uralitization	Pyroxene (Augite)	Actinolite / Hornblende	Fibrous green amphibole pseudomorphs after pyroxene prisms.	Hydration during cooling of gabbros, dolerites, diorites.
Chloritization	Biotite, Amphibole, Pyroxene	Chlorite (± Sphene)	Green pleochroic sheets with anomalous Prussian blue interference.	Late-stage hydrothermal alteration; retrograde greenschist.
Sericitization	Alkali Feldspar, Plagioclase	Sericite (Fine Muscovite)	Faint dusty brownish coating in PPL; bright speckles in XPL.	Hydrothermal wall-rock alteration (porphyry Cu deposits).
Serpentinization	Olivine, Orthopyroxene	Serpentine + Magnetite	Mesh texture (antigorite/lizardite nets around relict olivine).	Hydration of peridotites, ophiolites, ocean floor mantle.
Devitrification	Volcanic Glass	Quartz + Cristobalite + K-Feldspar	Spherulites with extinction crosses; microfelsitic chert texture.	Sub-solidus crystallization of volcanic rhyolites/obsidians.